

PROJECT REPORT

TITLE: Speech Enhancement using Noise Reduction in Python

SUBMITTED BY: Rajeswari

TOOLS: Google Colab, Python, librosa, soundfile, noisereduce

1. ABSTRACT

Audio recordings lo background noise valla voice clarity taggutundi. Ee project lo Python libraries use chesi noisy audio nunchi noise ni remove chesi clear voice unna audio ni create chesam.

2. OBJECTIVE

1. Noisy audio file lo unna background noise ni tagginchadam
2. Voice clarity ni penchadam
3. Original clean audio ki deggaraga enhanced audio create cheyadam

3. METHODOLOGY

Step 1: noisy_rajeshwari.wav.mp3 audio file ni Google Colab lo upload chesam

Step 2: librosa library tho audio ni load chesam

Step 3: noisereduce library use chesi noise reduction apply chesam

Step 4: Clean chesina audio ni enhanced_rajeshwari.wav ga save chesam

4. CODE USED

```
import librosa
```

```
import soundfile as sf
```

```
import noisereduce as nr
```

```
data, rate = librosa.load("noisy_rajeshwari.wav.mp3", sr=None)
```

```
reduced_noise = nr.reduce_noise(y=data, sr=rate)
```

```
sf.write("enhanced_rajeshwari.wav", reduced_noise, rate)
```

5. RESULTS

noisy_rajeshwari.wav.mp3 = Before - With Noise

enhanced_rajeshwari.wav = After - Noise Reduced

clean_rajeshwari.wav.mp3 = Reference Clean Audio

Observation: Enhanced file lo background noise chala taggindi.

6. CONCLUSION

noisereduce library use chesi manam noisy audio ni successfully

enhance chesam. Idi speech processing, podcast, call recordings lo use avutundi.